

JIA ZHANG



PRESENT APPOINTMENT

Research Assistant Professor

The Hong Kong Polytechnic University
Department of Computing & IHERD
Hong Kong SAR

Email:

joycezhangjia@gmail.com

Web: joycezhangjia.github.io

Phone: +852 2766 7348

RESEARCH INTERESTS

AI for Education • Computer Networks

Focus areas: Transport Protocols, Routing Security, and Low-Latency Services

EDUCATION

Tsinghua University | Beijing, China

Department of Computer Science and Technology

◇ **Ph.D.**, Computer Science, August 2023

– Dissertation: “Transport Layer Optimization for Mobile Web Services”

– Advisor: Prof. Mingwei Xu

– Thesis Committee: Prof. Jiahai Yang (chair), Prof. Ying Liu,
Prof. Wendong Wang (Beijing University of Posts and Telecommunications),
Prof. Shuhao Li (Zhongguancun Laboratory)

Tsinghua University | Beijing, China

Department of Electronic Engineering

◇ **B.Eng.**, Electronic Engineering, July 2018

– Second Bachelor's Degree: Business Administration from School of Business and Management

– Dissertation: “Implementation and Analysis of Congestion Control for the New Transport Protocol MPQUIC”

– Advisor: Prof. Mingwei Xu

PROFESSIONAL EXPERIENCE

The Hong Kong Polytechnic University | Hong Kong SAR

Department of Computing and Institute for Higher Education Research and Development (IHERD)

Research Assistant Professor

Sep 2026 – Present

Zhongguancun Laboratory | Beijing, China

Associate Research Professor

Feb 2026 – Aug 2026

Research Assistant Professor

Sep 2023 – Jan 2026

◇ Led independent and collaborative research on intelligent systems and network security

◇ Supervised teams and mentored students on research methodology and academic development

◇ Engaged in cross-institution collaborations with industry and academia to drive research

Tsinghua University | Beijing, China

Department of Computer Science and Technology

Research Assistant (Advisor: Prof. Mingwei Xu)

Sep 2018 – Aug 2023

Meituan | Beijing, China

Algorithm Strategy

Research Intern (Mentor: Mr. Xiaotian Li)

Sep 2021 – Jun 2023

Baidu | Beijing, China

BFE Team

Research Intern (Mentor: Mr. Sijie Yang)

Oct 2019 – Jun 2021

PROFESSIONAL
EXPERIENCE
(CONTINUED)

University of Surrey | Surrey, UK

5G/6G Innovation Centre

Research Assistant (Advisor: Prof. Ning Wang)

July – Sep, 2019

Tsinghua University | Beijing, China

Department of Electronic Engineering

Research Assistant (Advisor: Prof. Yongfeng Huang)

Sep 2016 – May 2017

SCHOLARLY
PUBLICATIONS

All publications are sorted in a reverse chronological order, where the # indicates that I am the corresponding author.

Refereed Papers in Conference Proceedings

- [C9] Yixuan Zhang, Shaorui Ren, Enhuan Dong, Haiping Wang, **Jia Zhang**, Zili Meng, Mingwei Xu, Shu Shi, Hebin Yu, Zhichen Xue, Yajie Peng, Xiaofei Pang, Jianping Wu, “Hermit: A Flow-Collaborative Transport Scheme for Multi-Source Video On-Demand Streaming,” accepted at the *IEEE International Conference on Communications (ICC’2026)*.
- [C8] Yixuan Zhang, Zili Meng, Enhuan Dong, Yan Zhang, **Jia Zhang**, Mingwei Xu, Jianping Wu, “Real-Time Video Gets a Fast Lane via Smart Queue Flushing at the Wireless Edge,” accepted at the *IEEE International Conference on Computer Communications (INFOCOM’2026)*.
- [C7] Shaorui Ren, **Jia Zhang**[#], Enhuan Dong, Mingwei Xu, Yixuan Zhang, Jiahao Cao, Jianping Wu, “Undermining Delay-based QUIC Congestion Control: A Receiver-driven Attack via Crafted Host Delays,” published in the *IEEE International Conference on Network Protocols (ICNP’2025)*.
- [C6] Andong Chen, Yangyang Wang, **Jia Zhang**[#], Mingwei Xu, “Right the Ship: Assessing the Legitimacy of Invalid Routes in RPKI,” published in *ACM Conference on Computer and Communications Security (CCS’2025)*.
- [C5] Peng Qian, Ning Wang, Foh Chuan Heng, **Jia Zhang**, Carl Udora, Rahim Tafazolli, “Enabling eBPF-Based Packet Duplication for Robust Volumetric Video Streaming,” published in *IEEE International Symposium on Computers and Communications (ISCC’2024)*.
- [C4] **Jia Zhang**, Haixuan Tong, Enhuan Dong, Xin Qian, Mingwei Xu, Xiaotian Li, Zili Meng, “Cold Start or Hot Start? Robust Slow Start in Congestion Control with A Priori Knowledge for Mobile Web Services,” published in *The Web Conference (WWW’2024)*.
- [C3] **Jia Zhang**, Yixuan Zhang, Enhuan Dong, Yan Zhang, Shaorui Ren, Zili Meng, Mingwei Xu, Xiaotian Li, Zongzhi Hou, Zhicheng Yang, Xiaoming Fu, “Bridging the Gap between QoE and QoS in Congestion Control: A Large-scale Mobile Web Service Perspective,” published in *USENIX Annual Technical Conference (ATC’2023)*.
- [C2] **Jia Zhang**, Enhuan Dong, Zili Meng, Yuan Yang, Mingwei Xu, Sijie Yang, Miao Zhang, Yang Yue, “WiseTrans: Adaptive Transport Protocol Selection for Mobile Web Service,” published in *The Web Conference (WWW’2021)*.
- [C1] Fangzhao Wu, **Jia Zhang**, Zhigang Yuan, Sixing Wu, Yongfeng Huang, and Jun Yan, “Sentence-level sentiment classification with weak supervision,” published in *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR’2017)*.

Refereed Journal Articles

- [J2] **Jia Zhang**, Shaorui Ren, Enhuan Dong, Zili Meng, Yuan Yang, Mingwei Xu, Sijie Yang, Miao Zhang, Yang Yue, “Reducing Mobile Web Latency Through Adaptively Selecting Transport Protocol,” *IEEE/ACM Transactions on Networking*, vol. 31, no. 5, pp. 2162-2177, Oct. 2023.
- [J1] Zili Meng, Yaning Guo, Yixin Shen, Jing Chen, Chao Zhou, Minhu Wang, **Jia Zhang**, Mingwei Xu, Chen Sun, Hongxin Hu, “Practically Deploying Heavyweight Adaptive Bitrate Algorithms With Teacher-Student Learning,” *IEEE/ACM Transactions on Networking*, vol. 29, no. 2, pp. 723-736, Apr. 2021.

PATENT	[P2] Mingwei Xu, Jia Zhang , Enhuan Dong, Zili Meng, Yuan Yang. Adaptive Transport Protocol Selection Method and Device for Mobile Web Service. CN112583818, Chinese Patent. Granted: Dec 24, 2021.
	[P1] Yangyang Wang, Andong Chen, Jia Zhang , Jiahao Cao, Mingwei Xu. Method and Device for BGP Route Origin Authentication Diagnosis Based on Feature Analysis. CN 118784362 A, Chinese Patent. Applied: Oct 15, 2024.
COMPETITIVE RESEARCH GRANTS	[G3] PI. Research on Decentralized Ultra-reliable and Faithful RPKI Routing Validation Architecture. CNY 300,000 NSFC Young Scientists Fund. No. 62502471 Jan 2026 – Dec 2028
	[G2] Co-I. Research on Key Technologies for IPv6 Network Security Risk Detection and Proactive Defense Total: CNY 3,000,000 MOST Key R&D Program of China. No. 2025YFE0201000 Nov 2025 – Oct 2027
	[G1] Co-I. Research and Validation on XXX Technology and its Application for National Internet Infrastructure. Total: CNY 480,000,000 National S&T Major Project on Cyberspace Security. No. 2025ZD1500100 Dec 2025 – Nov 2028
STANDARDS CONTRIBUTIONS	[S6] Jia Zhang , Yangyang Wang, Maria Matejka, and Mingwei Xu, “ASPA-based AS_PATH Verification for BGP Export,” <i>Internet-Draft draft-zhang-sidrops-aspa-egress-04</i> , <i>Internet Engineering Task Force</i> , Jan 2026. <i>Work in Progress</i> .
	[S5] Jia Zhang , Mingwei Xu, and Yangyang Wang, “Enhancing Route Origin Validation by Aggregating Validated ROA Payloads,” <i>Internet-Draft draft-zhang-sidrops-vrp-aggregation-04</i> , <i>Internet Engineering Task Force</i> , Feb 2026. <i>Work in Progress</i> .
	[S4] Jia Zhang , Mingwei Xu, Nan Geng, Chongfeng Xie, and Yangyang Wang, “RPKI-based Validation with Prioritized Resource Data,” <i>Internet-Draft draft-zhang-sidrops-prioritized-route-validation-00</i> , <i>Internet Engineering Task Force</i> , Oct 2025. <i>Work in Progress</i> .
	[S3] Jia Zhang , Shuhe Wang, Yangyang Wang, Mingwei Xu, Yu Fu, Wei Liang, Nan Geng, Jihua Zhou, Jing Gao, “Multi-source data provision system for RPKI route validation - general architecture,” <i>Communications Industry Standards of the People’s Republic of China</i> , Application No. H-202412190600. <i>Work in Progress</i> .
	[S2] Yonghong Fu, Mingwei Xu, Yangyang Wang, Jia Zhang , and Yuanyuan Zhang, “Operational Monitoring of RPKI Repositories Health and Safety,” <i>Internet-Draft draft-fu-sidrops-rpki-repositories-monitoring-00</i> , <i>Internet Engineering Task Force</i> , March 2026. <i>Work in Progress</i> .
	[S1] Shuhe Wang, Mingwei Xu, Yangyang Wang, and Jia Zhang , “Requirements for Monitoring RPKI-Related Processes on Routers Using BMP,” <i>Internet-Draft draft-wang-grow-bmp-rpki-mon-reqs-02</i> , <i>Internet Engineering Task Force</i> , Oct 2025. <i>Work in Progress</i> .
HONORS AND AWARDS	◇ Distinguished Contribution Team of Zhongguancun Laboratory 2025
	◇ Zhongguancun Laboratory Major Landmark Achievement 2024
	◇ Outstanding Contribution Team of Zhongguancun Laboratory 2024
	◇ Outstanding Communist Youth League Member of Tsinghua University 2018
	◇ Zheng Geru Scholarship 2017
	◇ Hongqian Electronics Scholarship 2016
PROFESSIONAL SERVICES	Conference Technical Program Committee Membership
	◇ ACM Workshop on Network and Operating Systems Support for Digital Audio and Video (NOSS-DAV). 2026.
	◇ IEEE/ACM International Symposium on Quality of Service (IWQoS). 2025, 2024.
	◇ International Conference on Big Data Computing and Communications (BIGCOM). 2025, 2026.

PROFESSIONAL SERVICES (CONTINUED)	Reviewer for Journal Manuscript Submissions	
	◇ IEEE Transactions on Parallel and Distributed Systems (TPDS)	
	◇ IEEE/ACM Transactions on Networking (ToN)	
	◇ IEEE Network	
	External Reviewer for Conference Manuscript Submissions	
	USENIX NSDI, ACM Multimedia, ACM CoNEXT, ACM HotNets, IEEE ICDCS	
	Other Organizing Services	
	◇ Student volunteer, ACM SIGCOMM 2019.	
TEACHING AND TUTORING ACTIVITIES	Teaching Assistant	
	◇ Computer Network Architecture, Tsinghua University	Fall, 2020
	Student Tutoring	
	◇ Postgraduate Tutor, Zhongguancun Laboratory	Fall 2024 – Aug 2026
	◇ Undergraduate Tutor, Tsinghua University	Fall, 2018 – Fall, 2020
MENTORSHIP	Students at Zhongguancun Laboratory	
	◇ Songlu Li. Ph.D. student (2026 –).	
	◇ Weihua Liu. Ph.D. student (2025 –).	
	◇ Jiayu Zheng. Ph.D. student (2025 –).	
	◇ Shaorui Ren. Ph.D. student (2023 –).	
	Students at Tsinghua University	
	◇ Yixuan Zhang. Ph.D. student (2020 –).	
	◇ Andong Chen. M.S. student (2022-2025). Now a professional engineer in Xi'an, China.	
	◇ Yan Zhang. Undergraduate student (2022 – 2024). Now M.Eng. student at Tsinghua University.	
	◇ Haixuan Tong. Undergraduate student (2022 – 2024). Now Ph.D. student at Tsinghua University.	
INDUSTRIAL IMPACT	◇ CERNET, China Telecom, and China Unicom use our RPKI data supplement system (CCS '25) for route validation.	
	◇ Meituan uses our congestion control selection mechanism, <i>Floo</i> (ATC '23), for mobile Web service (Dianping App).	
	◇ Baidu uses our protocol selection mechanism between TCP and QUIC, <i>WiseTrans</i> (WWW '21), for its video streaming service (Haokan Video).	
REFERENCES	Prof. Mingwei Xu	
	Professor, <i>Dean of</i> the Institute for Network Sciences and Cyberspace	
	Tsinghua University	
	xumw@tsinghua.edu.cn	
	Prof. Ning Wang	
	Professor, School of Electrical, Electronic and Mechanical Engineering	
	University of Bristol	
	n.wang@bristol.ac.uk	
	Dr. Nan Geng	
	Technical Leader, Data Communications Lab	
	Huawei	
	gengnan@huawei.com	